~~Holistic processing requires expertise, but expertise may develop rapidly~~

WHEN DOES EXPERTISE PRODUCE HOLISTIC PROCESSING, AND WHEN DOES IT FAIL?

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Background

Holistic processing is characterized by signature phenomena, including the disproportionate cost of inversion[1] and the difficulty of recognizing parts in isolation[2]. Although these effects were first identified in face perception, the mechanism that gives rise to them remains contested. One account attributes holistic processing to a dedicated **face template**[3]; another grounds it in long-term **perceptual expertise**[4].

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Face-like holistic processing in non-face stimuli

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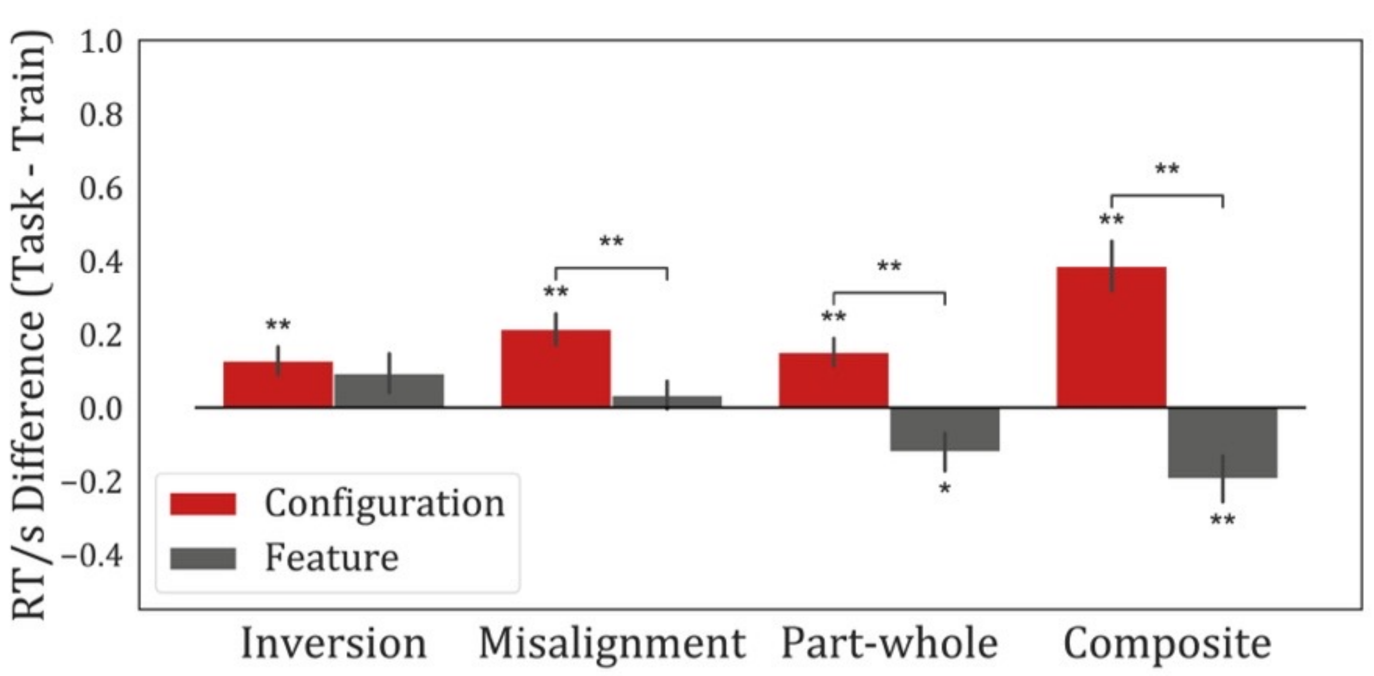
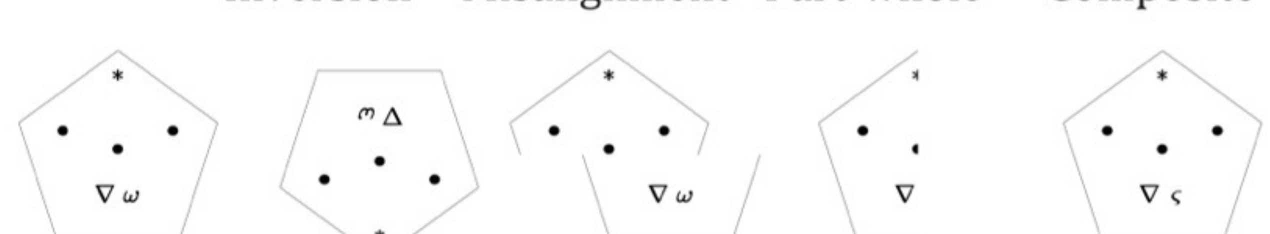
Our prior study, showed that holistic processing can emerge without a face template, driven solely by **configural information (the spatial relations among components)** rather than **featural information (the identity of individual components, independent of their spatial arrangement)**.





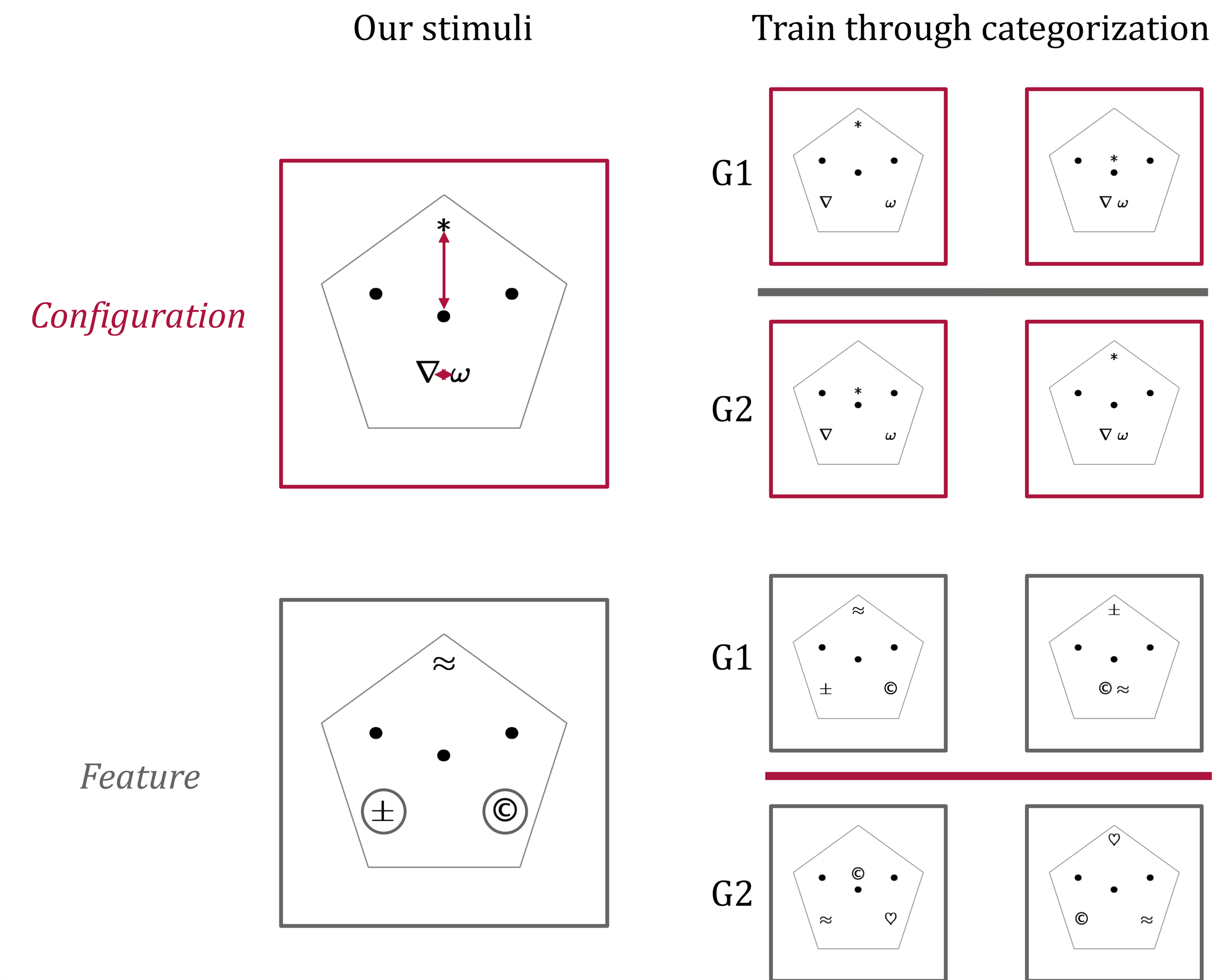
Critically, holistic effects appeared after only **~15 minutes of training (300 trials)** on configural information, but not on featural information. This leaves two open questions:

(1) Configural side: How does this rapidly-emerging holistic processing develop?

(2) Featural side: Would more extensive training elicit holistic processing?

Methods

We used the same non-face stimuli:



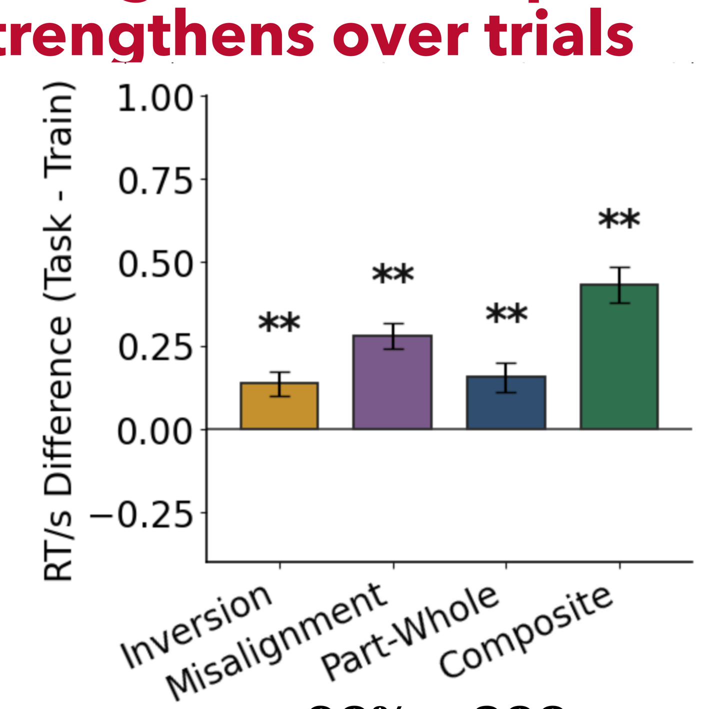
Experiment 1 (configural):

N=150, from OSU Research Experience Program (REP)
5 conditions (accuracy + trials): 90%+0; 90%+50; 90%+100; 90%+150; 90%+200. Each N = 30. (between subjects)

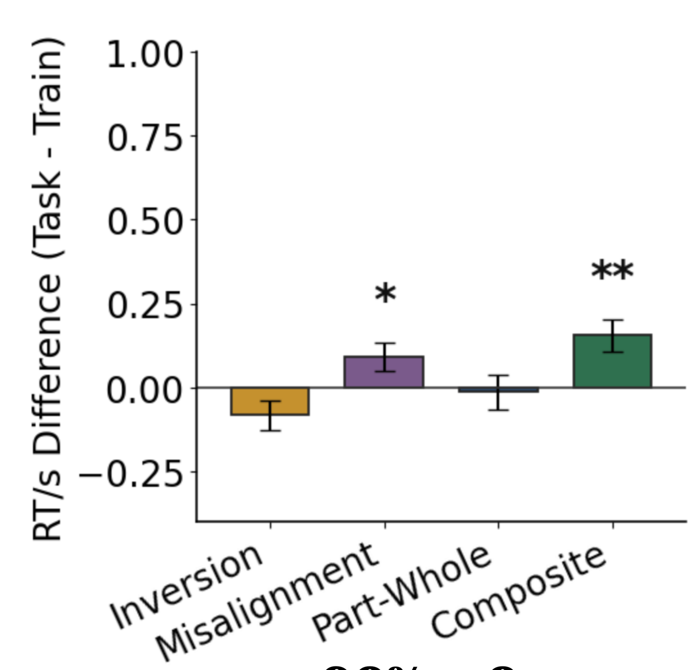
Experiment 2 (featural):

N = 5 (expect 20), from OSU REP
2 conditions: 90% accuracy + (0 - 200 trials); 90% + 8 hours training. (within subjects)

Configural holistic processing strengthens over trials



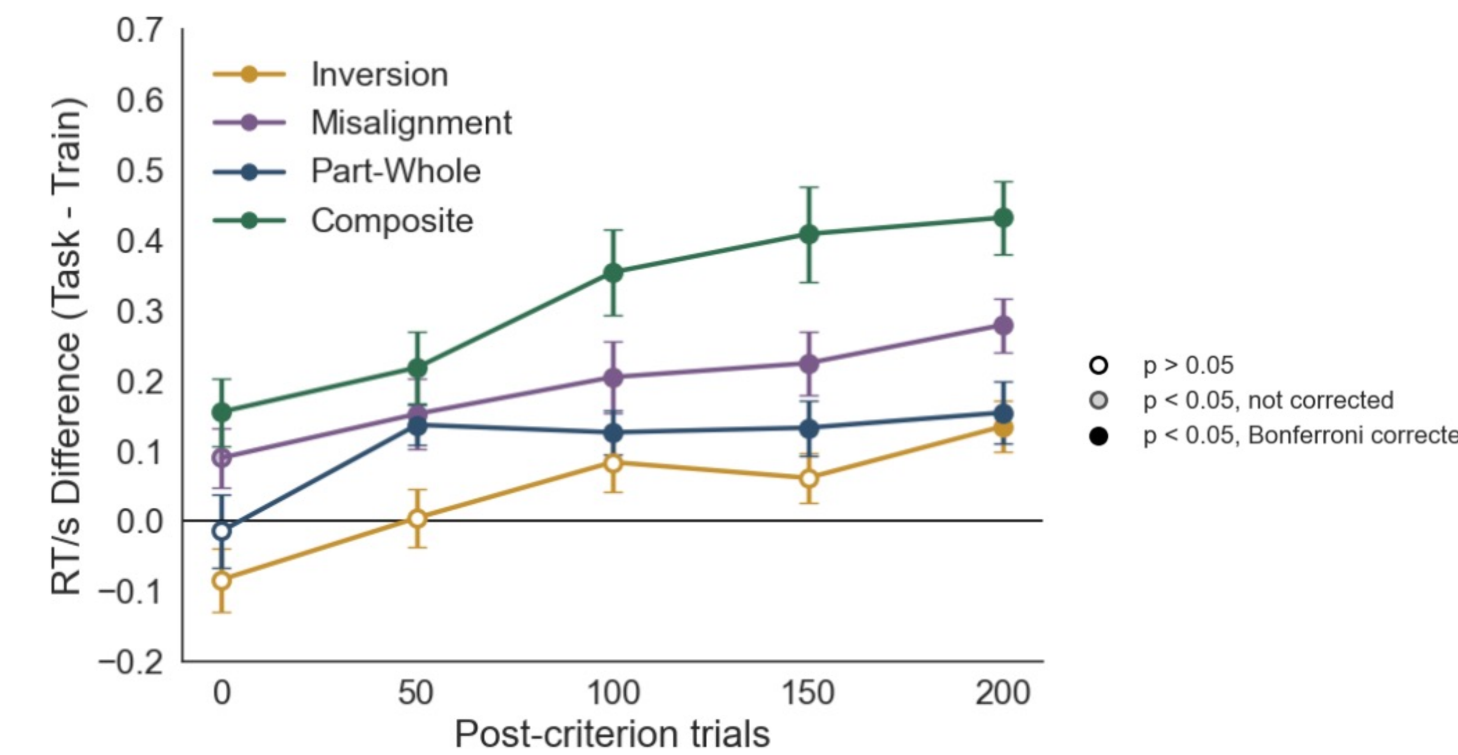
90% + 200



90% + 0

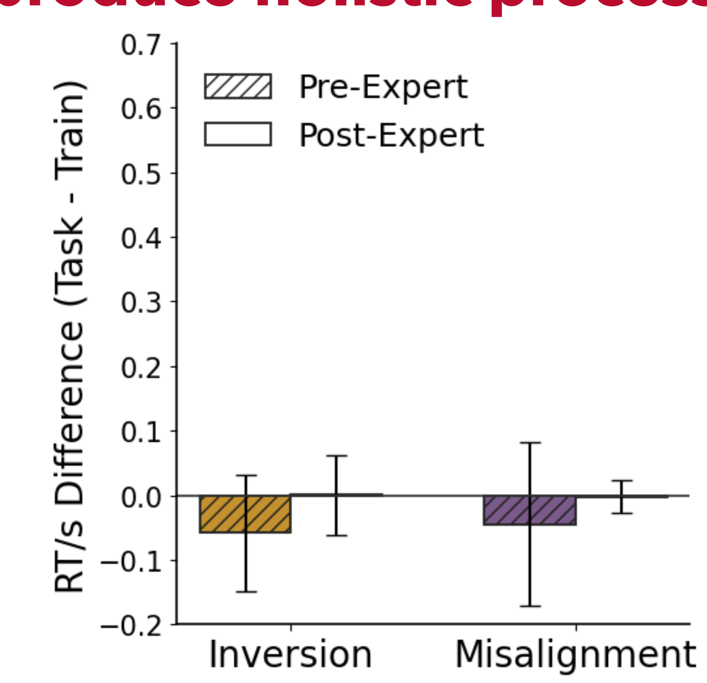
- 90% accuracy + 200 trials: We replicates the full pattern of holistic effects from our previous 300-trials study.
- 90% + 0 trials: A significant composite effect is already present, but not the inversion, misalignment, or part-whole effects.
- How do these effects emerge over the course of training?

Configural holistic processing strengthens over trials (cont.)



- Holistic effects differ in when they emerge, but not in how fast they grow.
- Linear regression showed no difference in slopes across effects, but significant differences in intercepts.

Featural training does not produce holistic processing



Even after ~8 hours of training **(14000+ trials)**, a duration typical of in-lab expertise studies[5,6], holistic processing did not emerge for featural condition

Conclusions

- **A clear dissociation: regardless of expertise, configural information elicits holistic processing while featural information does not.**
- Configural based holistic processing developed rapidly: composite effect is present from the outset, while other effect emerge within 200 trials.
- Featural based holistic processing does not emerge even after extensive training.

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